

University of Dhaka
Affiliated Engineering Colleges
B. Sc. in Computer Science & Engineering
2nd Year 1st Semester Final Examination-2021
EEE 2104 (Electronic Devices and Circuits)

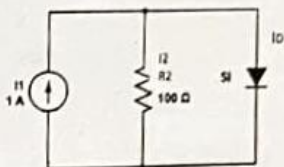
Time: 3.00 Hours

Marks: 70

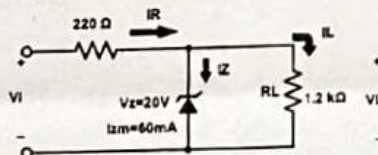
[Answer any Five questions]

1. a) What is semiconductor? Write down properties of semiconductor. 3
- b) Why we use semiconductor? Discuss about intrinsic semiconductor. 5
- c) What is LED? Explain working principle of LED. 4
- d) What do you mean by extrinsic semiconductor? 2

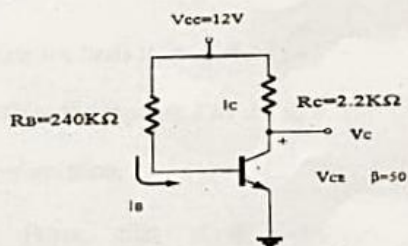
2. a) What is diode? Explain operating principle of bridge full wave rectifier. 5
- b) Discuss V-I characteristics of PN junction. 6
- c) Determine the currents I_Z and I_D for the network of Figure. 3



3. a) How Zener diode works as voltage regulator. Write down application of diode. 4
- b) What do you mean by clippers & clampers circuit? 2
- c) Write down advantage and disadvantage of full wave rectifier. 4
- d) For the Zener diode network of Figure. Determine V_L , V_R , I_Z , and P_Z . Where $V_i = 30$ V. 4



4. a) What do you mean by leakage current? 1
- b) Explain CE characteristics of transistor. 5
- c) Discuss about Transistor Circuit as an amplifier. 3
- d) Determine the following for the fixed-bias configuration of Figure. 5
 - (i) I_B and I_C .
 - (ii) VCE.
 - (iii) V_B and V_C .
 - (iv) VBC.



5. a) Write down the advantages of FET over BJT. 2
- b) Discuss working principle of JFET with necessary diagram. 7
- c) Explain the circuit operation of E-MOSFET according to Threshold voltage. 5

6. a) What is Operational Amplifier? Write the properties of it. 1+2
- b) For inverting Amplifier derive the equation given below, 7

$$A_{cl} = \frac{V_{out}}{V_{in}} = -\frac{R_f}{R_i}$$

- c) Find the output voltage for the circuit shown in Fig. 4

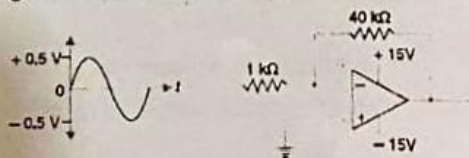


Fig.